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Developing a demand planning strategy for joint forecasting and employing analytical tool in an empirical case study

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Abstract
Product demand prediction is highly important as levels of inventory, customers forecast accuracy and overall performance are directly impacted by this demand planning. As the main problem of this research, accurate forecasting as a crucial challenge for achieving high performance is targeted. This paper has developed a demand planning framework with analytical capabilities of sales and consumer patterns, historical sales, and seasonality data to maximize the company's ability to satisfy consumer demand. The framework has been implemented in an empirical business environment and has been analyzed from a practical viewpoint. A comprehensive picture of all factors is acquired from this viewpoint for successful implementation. The primary objective of the framework is to enhance the demand planning process in joint forecast project and how these improvements will help other business units, a crucial component of the supply chain, to achieve greater performance levels. As a result, the process of demand planning is examined from a system viewpoint for its integration with the supply chain E2E process, performance, forecasting methods, and logistical structure which serve as the foundation for assessment of the current condition and context of the new system. A deeper focus is placed on the analysis of the demand planning processes and forecasting techniques. Several root causes are found based on the customer forecast inputs like forecast horizon, forecasting methods, standard process with customers, performance measurement and communication with customers. Integration of JDA, ERP, and other data sources, combined with flexible forecasting tools, greatly refines the demand planning while enhancing overall decision making. By blending statistical models with real-time customer insights, the firms maintain an agile, efficient supply chain that can quickly adapt to market disruptions.

Article Highlights

- Demand planning can be improved by integrating customer inputs into forecasting models.
- A robust analytical framework for consolidating JDA, ERP, and customer data to refine supply chain performance.
- Investigating through an empirical study the enhancement of automated forecasting tool accuracy, inventory alignment, and decision-making.
- Offering insights into refining joint forecasting by emphasizing flexible time fences.

Keywords Demand forecasting · Predictive analytics · Supply and demand · Supply chain management

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1 Introduction

Customer satisfaction and reliability are important elements in every aspect of business [1]. Demand planning examined as a part of the broader Sales and Operations Planning (S&OP) process is a vital part of the supply chain because it enables companies to satisfy consumer needs with sufficient inventory levels while keeping costs in check [2, 3]. Organizations may also improve customer satisfaction by including consumers in the process and considering their opinions [4]. However, while demand planning is a crucial component of every business strategy, it has faced challenges for its efficiency in empowering businesses in foreseeing and planning for future demand [5, 6]. This is due to reasons like the methods and systems that store the necessary information may not have all the features, which might make the job of demand planners challenging [7]. Planning systems may become the next problem impeding the S&OP process improvement after it has been successfully deployed in a corporation [8]. Demand planning methods usually must adapt to the limitations of information systems [9, 10]. Demand planning is challenging because of market uncertainty, but Artificial Intelligence (AI) can make it more effective [11]. These services are provided by a variety of solutions available on the market, but AI may be superior to them because it allows for better prediction at scale without bias or human mistakes. By forecasting future requests based on historical data and existing trends, AI can assist with demand planning [12, 13].

Moreover, today's dynamic business environment necessitates agile strategies to approach the complexities of supply chain [14], rapid market demand shifts [15] and environmental sustainability requirements [16]. Companies and their supply chain stakeholders have to face diverse product portfolios and geographically dispersed operations, which require sophisticated forecasting methods to remain competitive [17]. Emerging technological trends, such as AI-driven analytics for rapid product development [18], distributed ledger technologies for supply chain configuration [19] and effective production and inventory planning [20] hold the potential to strengthen supply chain resilience. Inaccurate forecasts lead to misaligned inventory levels and result in escalation of warehouse costs and customer dissatisfaction [8]. Consequently, improving the forecasting capabilities is inevitable for reducing risks and supporting the proper resource allocation and sustaining a competitive edge [3]. By integrating emerging technologies and more effective customer insights, this research aims to offer improved strategies for managing the complex challenges in demand planning and focused challenges like:

1. Poor visibility: Inability to perceive the big picture as demand planners cannot see everything. They can not take a big difference in market to foresee future trends because they can only observe what is occurring in their product sector [21, 22].
2. Poor forecasting: Forecasting mistakes by demand planners frequently lack precise knowledge of production capacity, or the amount of inventory required in the approaching seasons [6, 23].
3. Poor Consistency: Inconsistency of demand planners are unable to adhere to a standard demand planning procedure. Each business has its own set of procedures, which causes inconsistent data inputs and outputs along with complex analytics models [24, 25].

To fulfill the mentioned difficulties, the research studies suggest that the demand fulfillment must be data-driven and use big data technologies to be effective [3, 11, 13, 26]. The use of big data for demand fulfillment is one of the most current developments, which has been used in a variety of sectors [27]. Large volumes of consumer data may be analyzed using AI to produce insights that improve fulfillment. Moreover, S&OP and, more specifically, demand planning, are the challenges that many companies face [28]. The choice of information sources presents a typical problem to demand planning in a complex and global context where many businesses operate [29]. The planning process frequently involves a significant number of people, which poses difficulties for information flow, supporting procedures, and systems, as well as for the importance, accuracy, and applicability of the information, especially when it comes to a practical and operational business environment [30].

The evolving customer demands and the increasingly complex supply chain environment necessitate enhancements in forecasting processes [31]. This can be targeted by proposing a joint forecasting initiative that considers real-time customer inputs with automated predictive tools to reduce both the risk of overstock and stockouts while strengthening customer service levels. This requires improved data consolidation, heightened forecast accuracy, and scalable demand planning for improving the supply chain agility and long term competitiveness [32]. Standardizing forecast submission

timelines further streamlines operations, enabling demand planners to quickly identify and address deviations, thereby fostering greater collaboration and transparency with customers.

1.1 Problem description and research aims

The business case focused on in this paper has a structure for demand planning that uses a collaborative customer forecast with a time horizon of 3–6 months to increase the forecast accuracy. However, it is challenging to attain a higher fill rate due to the difference between forecast values and order quantities produced by the customers. The company must adhere to the customer-provided forecast numbers if the risks are greater, which eventually leads to a lower fill rate. In this case, responding to the customer demand will be a challenging task. Customers whose forecasted demand is too low but place high orders will cause the business to experience poor fill rates and excessive cuts, which will ultimately result in losses for both the company and the customer. Similarly, forecasting high demand and ordering low amounts leads to inventory costs [33]. The process of demand planning is meant to be the focus of this structure's perspective, while the Supply Chain and logistical organization are also considered. Technical, organizational, economic, and social factors will all be taken into consideration. In consideration of this, the following main question for this research is formulated:

- Examining the Root cause of the customer's orders that differs from what was forecasted, What is the impact of these root-causes?
- How the demand planning process can be improved to prevent the mentioned problems and enhance performance?

2 Literature review

2.1 Strategic planning and decision making

Strategic planning and decision-making are widely mentioned as some of the most crucial responsibilities of a company's board and management, but it has been discovered that senior management frequently lacks these skills. Some of these decisions might be related to long-term goals, while others might be related to short-term goals. These choices could be tied to short-term objectives in some cases and long-term objectives in others. The board must ensure that while making decisions, they are considering every aspect of the company. They must consider what is beneficial to the company as a whole and not just for one division or department [34].

Strategic decisions can be accomplished in a variety of ways and contexts. It consists of:

- Analytical method
- An intuitive strategy
- Participatory methodology
- Hierarchical methodology

Demand forecasters create predictions about future consumer demand, keep tabs on historical demand and inventory levels, examine sales orders and marketing strategies to anticipate customer needs, spot potential production bottlenecks, and suggest operational adjustments to boost efficiency. Demand planners frequently must strike a compromise between short-term needs and long-term resource allocation estimates. A company's decision-making will also be influenced by the kinds of items it sells. For instance, a company's decision-making would be more centered on lowering expenses than increasing profit margins if it had a large mix of low-margin and high-margin products.

Along with handling international risks like currency fluctuations or political unpredictability, demand planners need to find new markets and chances to increase sales. Software for demand planning may be employed as a strong decision-making tool in this situation since it can not only improve forecast accuracy but also provide information on potential external influences on future activities like revenue growth or customer satisfaction. The use of AI technology in demand planning is becoming more common due to its precise forecasts that assist businesses in making decisions devoid of human bias. AI employs data analytics algorithms that can accurately and precisely anticipate future events based on historical trends, something that is impossible for humans to achieve [33].

2.2 Demand planning as a component of strategic planning

Companies estimate the demand for their products using the demand planning process. It is based on market information that is currently accessible and previous data [17]. Demand management is a practical strategy for adapting to shifting market conditions. The ability to estimate product demand and production gives managers the knowledge they need to make more informed decisions in the future. Customers and businesses may both benefit from demand planning, which has a range of advantages such as increased profitability, customer happiness, and corporate efficiency [21]. The warehouse's inventory levels will be influenced by the organization's demand strategy, which will also have an impact on service levels [35]. This can be an issue since it becomes challenging to make precise estimates for future demand when demand levels are changing quickly. There are five primary parts:

- Prediction
- Improvement
- Sourcing
- Selection
- Review

Demand planning forecasting involves making decisions now as well as in the future to determine what adjustments are necessary to make the operations and capital expenditures for future production [36]. The cost of sourcing and selecting operations, along with how much inventory should be created or bought at certain points in time, as well as when production capacity will become available, are all factors that are considered when forecasting. The level of optimization might be global or item-level, and it may also entail modifying either external sources (like suppliers) or internal processes (e.g., R&D). Finding possible suppliers that can fulfill the requirements from among those considered suitable based on certain criteria is considered sourcing [37].

2.3 Sales and operations planning (S&OP)

S&OP has been developed to meet the requirement of bringing strategy to the everyday life of businesses and to allow cross-organizational collaboration as the topic of strategic decision making has shown to be a frequent topic in organizational study. S&OP's primary goals are to balance supply and demand as well as create connections between a company's strategic and operational initiatives. The dedication of the participants is one of the crucial success factors of effective S&OP. By improving the process for giving stakeholders the essential information, this type of commitment can be formed [38]. The key emphasis for businesses that prioritize service differentiation is the relationship between customer demands, customer centrality, innovation, and service differentiation. Customer demands are not a significant predictor; however, customer centrality is a strong predictor of service differentiation and innovation. Additionally, it has been discovered that the complexity of consumer wants is favorably correlated with service distinctiveness but negatively correlated with innovation [39].

The traditional view of S&OP is that it's a process which is completed monthly. It is important to do this to review the company's performance and make any necessary adjustments. However, there are many ways to complete this process, and it should be tailored to fit the needs of the company. The demand S&OP puts a strong emphasis on customer demand and aids in matching the supply chain to it. The sales and operations teams collaborate in this process to estimate demand, schedule production, and adjust inventory levels as necessary. The major objective of this procedure is to reduce lead times and costs while maintaining customer service standards to optimize inventory levels [40]. The process of S&OP is often seen as a five-step process, as illustrated in Fig. 1:

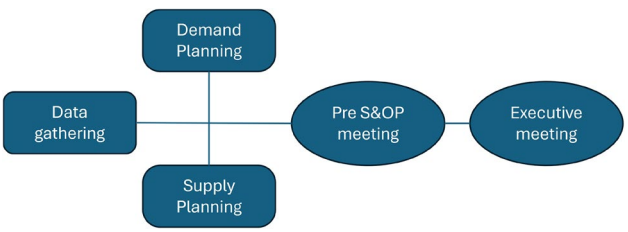


Fig. 1 S&OP Process

1. *Data gathering:* The data that is obtained in S&OP is essential in demand planning. The information should be gathered from many sources and used to spot patterns and trends throughout time. It is crucial that this data not only be gathered but also properly exploited to impact the plan. Numerous methods, including as market research, consumer feedback, data analytics, and competitor analysis, can be used to collect the data [33].
2. *Demand Planning:* The process includes cross-functional cooperation. It is regarded as one of the S&OP process most unpleasant elements. One technique employed by senior management to comprehend the effects of supply chain-side volatility and demand uncertainty is scenario analysis [41].
3. *Supply Planning:* The capacity aspect of any company is assessed through supply planning. Supply planners must consider many aspects to make efficient plans. This involves planning for operations, materials, and manufacture. It is up to supply planners to decide how much product to produce, when to produce it, and where to store it [42].
4. *Pre-S&OP:* During the pre-S&OP meeting, all department heads come together to assess the planning and, where possible, execute. During this conference, the financial effects of demand and supply chain planning are being considered. The traditional strategy is an event-driven, reactive procedure. The more contemporary technique is a proactive procedure that is planned and carried out to foresee opportunities and act on them [5].
5. *Executive meeting:* In some cases, the pre-S&OP meeting cannot resolve all the difficulties. To approve final decisions and the overall S&OP plan, executives consider the department heads' proposals during this meeting [43].

2.4 Conclusion from literature review

One of the essential components of an effective planning process is the demand planning strategy. Although earlier industries frequently worked more in a supply-driven manner, the market is shifting to become more customer-oriented. Customers are in the driving seat because the forecast of the products is determined to meet their demands. This is accurate across many sectors that had an impact on how business operations are structured. To be able to meet the market's ever-shifting demands and to take advantage of its opportunities, planning and forecasting future demand is essential. Highly accurate predictions and plans may aid businesses in overcoming difficulties such as longer lead times for materials to be procured, planning the product range, capacity planning, and providing suppliers with more precise estimates to prevent the supply chain bullwhip effect. Due to this, each process should be designed exclusively to meet the company's demand. Along with the degree of plans and the frequency of planning, the schedule for the plan should also be considered. The planning schedule is frequently determined by the companies. The timescale must be close to two years or more to take strategic decisions into account along with tactical ones. This aids in planning the demand for goods by considering the technological product frameworks. It is usually not practical to formulate demand plans at the Stock Keeping Unit (SKU) level since product portfolios are getting complicated and the portfolio will be changing. A statistical forecast based on historical data is helpful to aggregate and disaggregate at multiple levels to forecast the future demand. This can involve financial controllers reviewing the sales of a specific business unit or product managers considering global data per product line, sales managers finding the sales of a specific region, operations managers reviewing the planned demand for a specific factory. This allows an area manager, for instance, to modify the plan at the country level and have it implemented at all levels where the business is booming. This enables those engaged to see the wider picture and lessens the need for highly developed analytical abilities throughout the forecasting process.

The changes in consumer behavior, the changing nature of the market, the expansion of market share, and various types of project information are among the potential judgment adjustments that cannot be observed from historical data. Most of the research studies have discovered that judgment modifications made to data baselines frequently boost prediction accuracy, particularly when the adjustment is larger and not simply a minor adjustment based. The projections are frequently biased by the judgmental modifications, and the projected statistics frequently exceed the actual. The preconception is reduced to determine the change in plan. The planning process should be in place before deploying planning systems. To facilitate the sharing of data and the development of the plans, the demand planning process and systems should be developed. Some of these systems are straightforward databases or software applications. These software make a forecast of future demand based on current data from markets and consumers using analytics technologies. Companies frequently commit the mistake of failing to have the proper kind of system in place that will facilitate this process. This may cause inefficiencies in forecasting the demand, which would fail to meet the customer's demand.

3 Demand planning strategy for joint forecasting and employing analytical tool

3.1 Business demand planning and logistical structure

The Business domain in this study includes a multinational company. It manufactures and develops products, and sells & distributes Fast-Moving Consumer Goods (FMCG) products in beauty industry. It has customers in numerous industry sectors. Its regional commercial business units serve as the foundation for the corporate global market reporting structure. It offers products to Business-to Business (B2B). The business is growing to the position of market leader. The business units focus on two categories of consumer beauty and luxury products. The production of business firm products and regular business activities are achieved through their supply chain. For the benefit of consumers, they produce, market, sell, and distribute a variety of products. They still rely on a global network of Third-party Manufacturer (TPM), who create about one fifth of the final products. Additionally, they buy packaging parts that are produced in accordance with their design requirements. Raw materials include a higher risk of labor and human rights abuses, and it is considered to enable reduction of this risk. As a large international company, it also engages in marketing, offices, retail locations, and internet, among other supply chain activities. Based on supplier documentation pertaining to the following four major categories and evaluating the environmental and social performance of their suppliers: Sustainable Procurement, Labor and Human Rights, Ethics, and Environment.

3.1.1 Application of Tools for demand planning

1. *JDA:* James Donald Armstrong (JDA) is a statistical forecast-based system. Everything which is forecasted by the demand planners will be in this system. JDA is linked to the production system but not an end-to-end system. With a few exclusions, the studied company will position its forecast in JDA globally. The statistical recommendations will be used to compute the regular products. Statistical forecast is calculated based on at least 12 months history. The forecast will be more precise as there is more historical data. However, it can have a negative effect in specific situations, such as Covid. Every initiative must be placed in JDA. Whatever is placed in JDA, it is subsequently used by other side of supply chain such as factory planners for production. The data input aggregates data from various sources, including JDA for forecasting and ERP for inventory management. It consolidates historical sales data, customer forecasts, and relevant market data to create a comprehensive dataset. In the processing layer, the data undergoes transformation and cleansing to ensure quality and consistency. This process involves standardizing formats, addressing missing values, and applying relevant business rules.
2. *MFR:* Manugistics Forecast Reporting (MFR) is a reporting system. Demand planners can extract the data from JDA and from ERP. The latest Forecast Performance updates in a MFR system. The analytical tool is designed with a modular architecture. The data input aggregates data from various sources, including JDA for forecasting and ERP for inventory management. It consolidates historical sales data, customer forecasts, and relevant market data to create a comprehensive dataset. In the processing layer, the data undergoes transformation and cleansing to ensure quality and consistency. This process involves standardizing formats, addressing missing values, and applying relevant business rules. The core of the tool, this engine utilizes various forecasting algorithms, including: Statistical Models which are traditional time series analysis methods, such as the Lewandowski model, which consider trends, seasonality, and historical patterns. The results are presented in an intuitive dashboard format, allowing users to visualize forecast accuracy, track performance metrics (such as Mean Absolute Percentage Error 90 (MAPE90)), and assess deviations in real-time.
3. *Prometheus©:* The tool for producing content is called Prometheus©. It employs AI technology to create material for themes, and in areas where it falls short, it may use an automated bidding mechanism to select the best-fitting freelancer. In the studied company, Prometheus© is an ordering system for all initiatives for luxury brands. It provides offerings through Electronic Marketing Book (eMB).
4. *ERP:* It is a connecting tool to a company as well as to the Customers. The demand planners and customers can enter their inputs in this system.

3.1.2 Markets and logistical model in EMEA region

The studied company operates in markets with intense rivalry and high customer demand. There is competition in the market and variable customer demand because of the nature of the generic products which are manufactured. The EMEA region includes several locations. The company has a strong regional focus. Therefore, only the European distributor market is focused. A monthly local planning process is used for this forecasting, and several stakeholders participate by contributing to the demand plan individually. Every business unit has different tools depending on the requirement. While European Distributor Market (EDM) and Germany, Austria, and Switzerland (DACH) markets both have unique tools, they are both part of the EMEA region. Because EDM works with distributors and the distributors then sell the products to retailers, whereas in DACH and the UK, it directly sells the products to retailers. It is always necessary to find the match for the system requirements.

The logistics model can be characterized as improving forecasts using the appropriate methods. Reduced expenses, increased margins, and improved cash flow are all part of the corporate plan. In this business sector, cost reduction and improved customer satisfaction are the main objectives of logistics. Flexibility and a shorter lead time are usually secondary objectives. Deliverability dependability is a key objective. This shows that cutting costs and upholding service levels are at odds. The demand planning procedure involves collecting joint forecasts from preferred customers and determining if every customer is abiding by the studied company's rules and regulations to sell their goods in the authorized area. Demand planning is based on forecasts and is divided into planning and execution parts. Historical sales data, market hypotheses, and statistical predictions are all employed as data sources. Demand planning (EMEA and corporate) and monthly S&OP are responsible for reviewing. Through ERP systems, information is communicated in EDM.

The geographical location of the logistics system includes north of Europe for the central team and marketplaces for the regional teams. The global Distribution Center (DC) network for the studied company is huge worldwide. Main factories in Europe where the items are made are in EU and UK. The flow of goods occurs from Suppliers to factories and DC, and then to Customers and Direct to Consumer (DTC). Resale Price Maintenance (RPM) and marketing suppliers are one of the suppliers, and the goods movements occur straight from the manufacturers to the DCs, TPM External Operation Planning (EXO), and ultimately the customers. The flow of products from DCs to various business units and suppliers and the product flow within the studied company as internal flow of are illustrated in Fig. 2.

3.2 Proposed demand planing process

Demand Planning can be defined as the art and science of planning customer demand to drive holistic execution of such demand by corporate supply chain and business management. Starting with a reliable statistical prediction, demand planning adds intelligence through consensus management of the demand information process. Demand Planners are subjected to forecast several things, including:

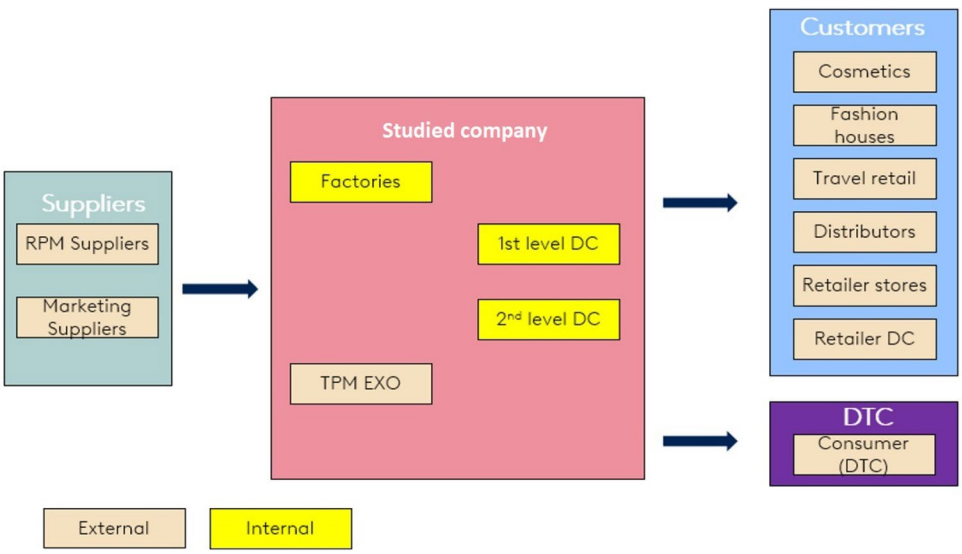


Fig. 2 The studied company internal and external Product flows through the global supply chain

1. Shipments to Customers where returns are not included in the forecast or shipments.
2. Units which are converted to a corresponding output of gross sales.
3. A forecast horizon of minimum 18 months.
4. Unconstrained demand unless there is a supply issue communicated through hubs/division.

Forecast is done to meet the need of Supply chain which involves deployment to DC, production at Factory, and capacity planning at DC/factory. It also includes commercial partners which involve S&OP and financial Planning. The forecast is done by creating a baseline forecast using historical shipments and then adding market intelligence from sales and marketing plans, promotional plans, and feedback & discussion from the S&OP. Demand forecasters don't use the forecast for:

1. Managing stock levels
2. Using the forecast as a contingency plan
3. Forced to equal target or objective
4. Manipulation (i.e., no change or intentionally high or low)
5. Negotiation (i.e., low forecast until more funding provided)

The role of the demand planner is to minimize the inaccuracy with a focus on continuous improvement. They balance the customer's requirements with the capabilities of the supply chain, enabling them to match supply with demand proactively and execute the plan with minimal disruptions and cost. A forecast is created from the bottom up at Demand Forecasting Unit (DFU) level. A DFU is a combination of Demand Location (Market) and Demand Group (subset if required). The forecast is built using statistical methods incorporating business intelligence assumptions collected from Marketing and Sales. Demand planners comprehend macro market trends within a market. They are noticing significant changes in consumer behavior and major changes in market size. Through cooperation with the business partners, they can comprehend the connection between market size, shipments, and share. They understand that the biggest causes of forecast bias in businesses are internal work procedures and macro events.

3.2.1 End to End (E2E) Supply chain

From product creation through distribution, the studied company's worldwide supply chain oversees the production and sale of FMCG in more than 100 countries. The studied company has changed every aspect of its end-to-end supply chain to provide consumers with innovation. It has employed end-to-end efforts such as big data platforms, data analytics models, and product design optimization techniques to satisfy customers' requests. Data analytics, robots, data science, and other technologies are being used by the corporation to develop across the whole lifespan of its goods. The E2E supply chain includes two main divisions, such as business units and hubs. The master data includes DRP calculations considering demand and existing stock, such as the requirement by location based on parameters. The Site Integrated Planning (SIP) includes MSP/EXO/MSP for the generation of production and materials supply plans for the plants, including forward forecast and

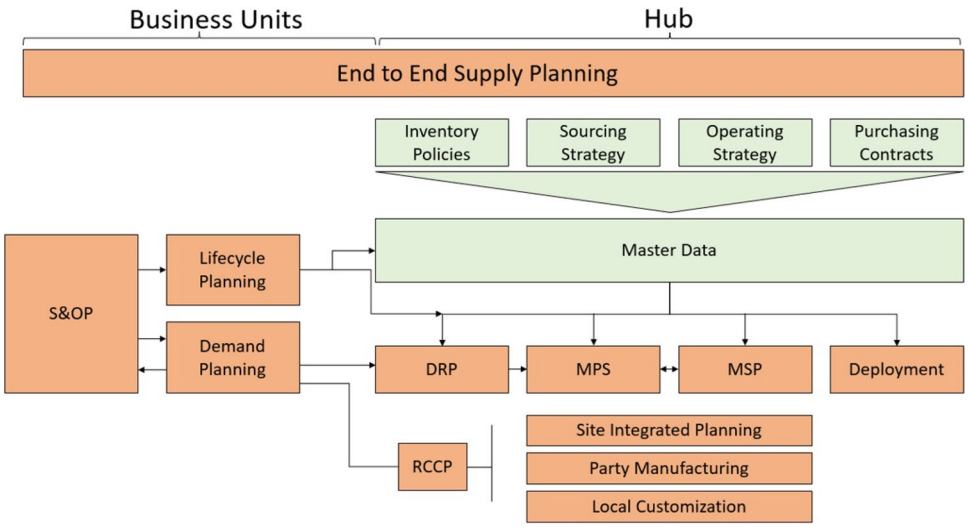


Fig. 3 The studied company Supply Chain structure across the business

capacity plan (in-house and supplier) as in Fig. 3. Deployment is a constrained plan for execution of what to supply, based on available stock. Life cycle management team is the upstream representative of Planning who influences the business. Life cycle management integrates people, data, processes, and business systems to create value at the different stages of the life cycle.

3.2.2 The studied company's master data architecture

Master data must reflect business strategies agreed upon with internal and external stakeholders, as well as ERP system functionality. On a regular basis, Master Data must be reviewed for accuracy and any changes in business strategy, including Phase In-Phase Out decisions. The master data is stored in different tools:

1. Master data is maintained in ERP and daily transfer occurs from it to JDA. The master data transfer is:

    (a) Finish products: Demand units/International Article Number (IRC)
    (b) Global materials Master (GMM) Hierarchies: Brand Hierarchy & Product Hierarchy
    (c) Product Attributes: Size, Article Type, Gender, etc.

2. Master data is maintained directly in JDA, DFU with effective and discontinuation dates and other master data fields that can be populated in JDA (Base Price, PIPO dates)
3. Other master data (considered as configuration) maintained in JDA by the support team

The JDA tool consists of Demand Unit, Demand group, Location, and Model to adjust the forecast. The DFU is the basic unit of demand, and it is at the lowest level in JDA. DFU is the level at which a forecast is created and stored (for example: statistical forecast, promotion). DFUs are typically generated automatically following the GMM interface; if not, Demand Planner (DP) can create. Demand Unit is the material code or IRC. They are interfaced from GMM and visible in the Demand unit table. Brand hierarchy, Product hierarchy, and attributes are assigned to this object, also via GMM interface.

3.2.3 Hierarchies of demand planning

There are three different hierarchies for IRC.

1. Product hierarchy occurs at 3 levels: It represents to which line, brand, house or division the product belongs to
2. Market category happens in 4 levels: Core category, subcategory, market category, and product segment. It represents the category of products
3. Article type occurs at 2 levels: Article type and article sub type. It represents how the products are sold (i.e., one shot/Promo, shopping bag etc.) and the type of product (i.e., Merchandising, sample etc.)

3.2.4 Distribution requirements planning (DRP)

DRP is a systematic process to determine which goods are required, in what quantity, at which location, and when they are required. The purpose is to meet demand within the distribution network according to desired coverage parameters. The process takes place in JDA. For each location in the distribution network, the DRP calculation takes into account the inventory, demand, and parameters to determine the quantities required to be received from the upstream locations, and the quantities required to be sent to any downstream locations. This is the Unconstrained Distribution Plan as

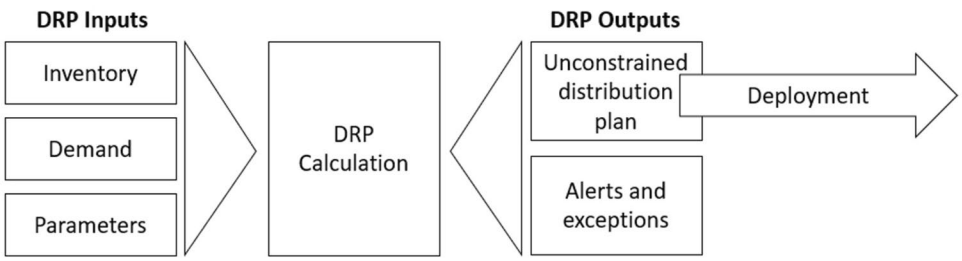


Fig. 4 The studied company's Distribution requirements planning process

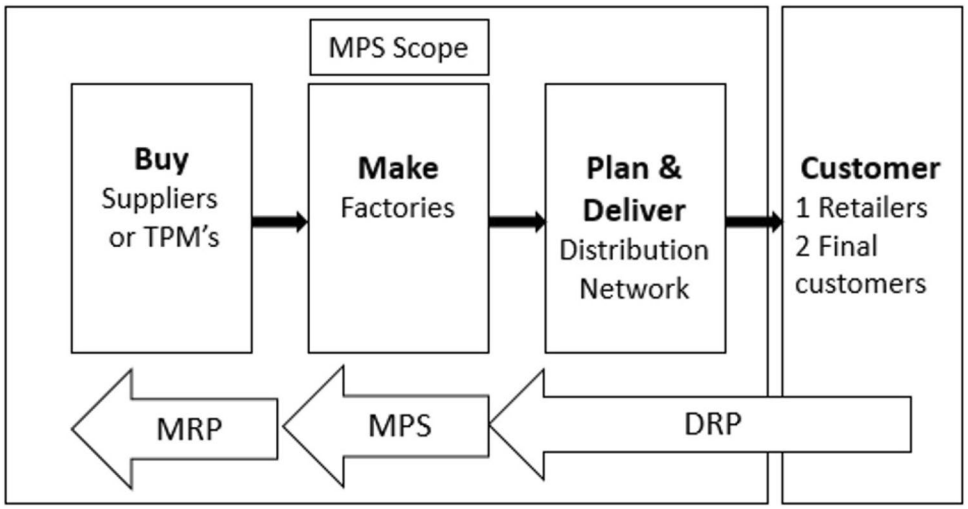


Fig. 5 The Master Production Schedule (MPS) process

illustrated in Fig. 4. To ensure that the DRP calculation is as accurate as possible, and delivers the required business outcomes, parameters must be regularly reviewed and maintained. Network Parameters represent real-world aspects such as loading and transport times, so these are compared vs actual times and adjusted as required. SKU Parameters are reviewed in relation to desired business performance, to optimize service and inventory. This is supported by ABC review, the process of segmenting the product portfolio by importance (sales value) and defining parameters to reflect this. The Deployment process in Fig. 4 describes the generation and execution of a constrained distribution plan to supply the whole distribution network. The aim is to ensure product availability for sales in all markets to maximize service, and to optimize inventory and distribution costs. The outcome of the deployment calculation is a feasible distribution plan to be analyzed and executed.

3.2.5 MPS and MSP

The MPS process describes the generation of an executable constrained production/supply plan at a production plant, with the aim of achieving customer service objectives while optimizing cost and inventory performance. The main principles of MPS as illustrated in Fig. 5 are:

- Resolve material constraints by prioritization.
- Give visibility and optimize availability to meet downstream needs.
- Provide capacity figures to allow resource planning.
- Use of Planned Time Fence: firmed production

MSP is considered as the process to ensure the availability for externally procured materials for use in the production of finished goods, to support service to the customers, while optimizing cost and inventory performance as illustrated in Fig. 6. Material Requirements Planning is the system calculation of a recommended material supply plan and the dates on which the purchased items should be received. The main principles of MPS are:

- Acting immediately on constraints and abnormal situations: solve, collaborate, escalate, prioritize.
- Data in the system must be up-to-date and realistic, for use by downstream processes.
- Collaborate with suppliers. Work to understand their operation and their supply chain; understand and analyze their capacity; set service level agreements targeting responsiveness; challenge them to be responsive.
- Drive manual work out of Material Supply Planning. Engage the principle "silence is approval" with suppliers. Use collaboration platforms such as e2open, targeting vendor managed inventory.

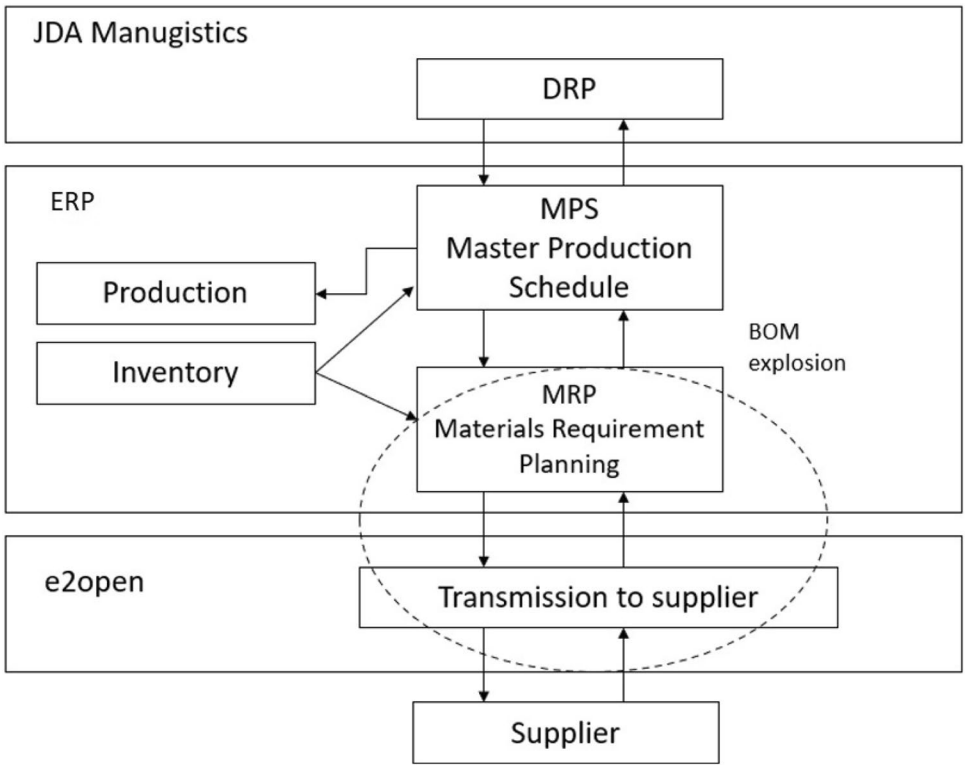


Fig. 6 The studied company's Material Supply Planning (MSP) process

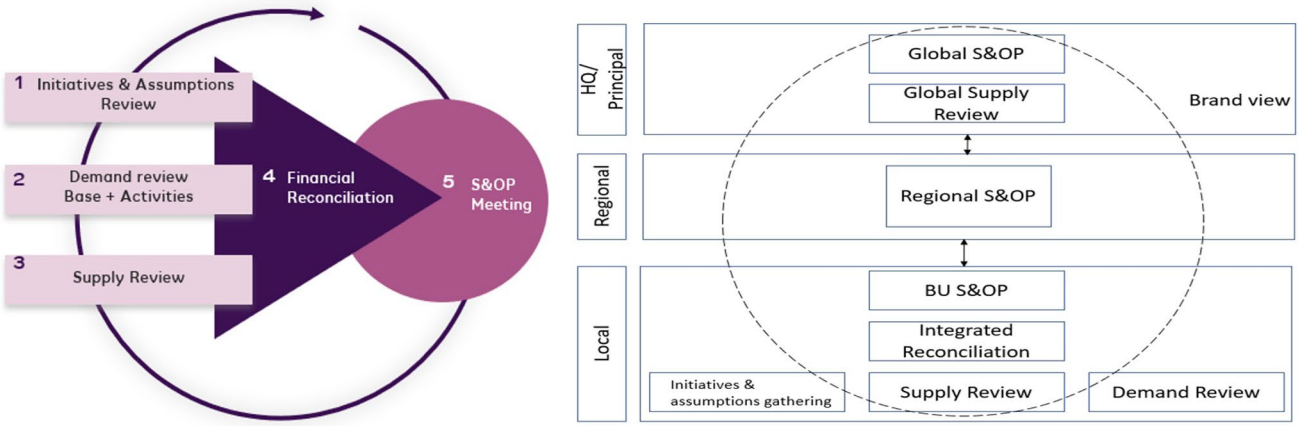


Fig. 7 The studied company's Sales and Operations Planning (S&OP) process and outline of the global S&OP Governance

3.2.6 External operation planning (EXO)

EXO is considered for the planning of finished goods manufactured externally at TPMs, as opposed to factories owned and operated. The use of a TPM relates to a strategic choice not to manufacture in-house, usually related to: A niche technology with specific manufacturing technologies which does not make sense to invest in yet, A technology which the company has chosen not to be a specialist in manufacturing, An area where the studied company is not cost competitive vs TPMs, or where there is no tendency for managing the complexity, and a formula or product design owned by a TPM, which the company wishes to bring to market.

3.3 S&OP process at the studied company

The proposed S&OP cycle as illustrated in Fig. 7 consists of five steps:

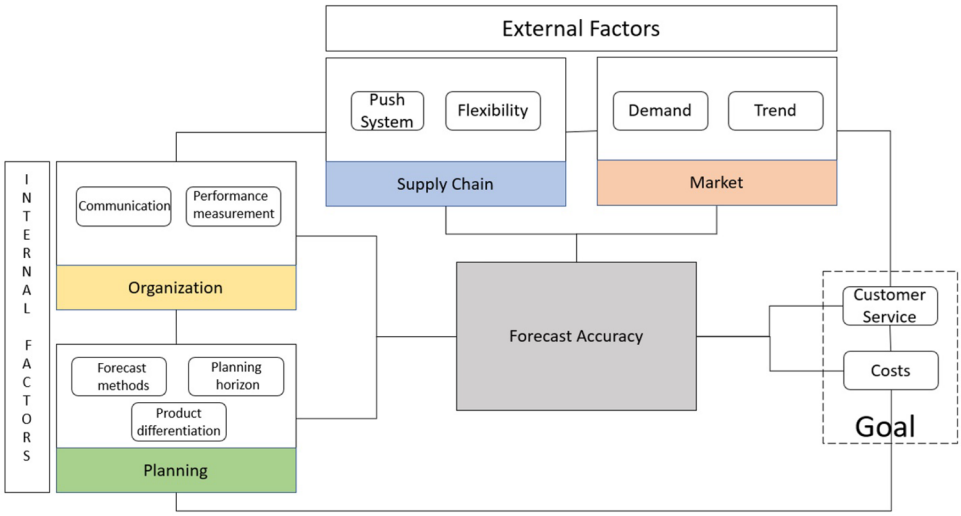


Fig. 8 The flow diagram model for overall root cause

1. Baseline: Establishing company targets and comparing projected results to those targets.
2. Initiatives and assumptions: Initiative master plan is created by agreeing on, validating, and sizing/phasing new tasks. It is important to connect assumptions about market mechanics, such as distribution routes and market dynamics, with promotions and other events and establishment of forecast and execute plan.
3. Demand review: The first step in the demand evaluation process is to provide a precise prediction of the goods and services that consumers may require. Knowing what resources will be required to produce these products or services is crucial. When both of those items are there, planning can begin on how to swiftly put them together for what consumers will need in the present and the future.
4. Supply review consisting: review of critical resources, deploying alignment of sales and resources, analyzing supply chain performance- cash, cost, service
5. Integrated reconciliation: Gaps that need to be filled with action should be identified and emphasized. Scenarios are developed to be utilized in S&OP decision-making.
6. S&OP: Reviewing potential critical gaps scenarios and deciding on a course of action. Deciding on sign-offs and Send the approved decisions back for implementation.

3.4 Integrating demand forecasting

The forecasting techniques employed for the product groupings in this research are examined to increase forecast accuracy and create a better demand plan. It is helpful to get an understanding of the impact of a forecast adjustment before assessing the impact of other forecasting techniques. If the prediction is more accurate, it will be easier to determine the proper number of items. A forecast's total number of products and degree of inaccuracy are connected. The forecasted quantity of goods will increase if the inaccuracy diminishes, as it should. The primary focus of this study is on strengthening the brand-new project process and strategy that was just launched in the studied company in the project named joint forecast with the assistance of consumers. This section will focus on the enhancements made to the joint forecast project process. The section will be elongated in the forecasting methods called Joint forecast with customer collaboration. It will also cover the difficulties the demand planners first encountered and how those challenges were resolved. In the next section, it will also include what other improvements can be made to make the process even better.

The complete root cause is depicted in Fig. 8. Both the organization and technique, as well as the planning quality, are considered. The elements in this graphic that directly affect forecast accuracy are included. These so-called root causes are connected to both the organizational element and the standard of the planning. The flow diagram is at a generic level, and it is difficult to understand the precise linkages. A so-called issue tree can be used to describe the situation and help address it more effectively. An entire solution will emerge by addressing the root causes, which are the problems at the base of the issue tree. It is explained where the procedure's primary uncertainties are and what their primary causes are involved in the joint forecast project process. The forecast accuracy is the main element utilized in this aspect. Performance can be improved by looking at the uncertainties and enhancements for this indicator.

Several forecast techniques are used in the studied company, and some of them are discussed in this section.

Benchmarking - No Historical data for new product launches: Currently, qualitative methodologies are generally used in the forecasting process for new launches of products. Product launches may be utilized to advertise the newest and best products in the range and genuinely put the business on the map. Before developing a new product, it is commonly recognized practice to consider what has been successful in the past. To inform next launches, there is not any information about previous launches, nevertheless. Because of this, it is challenging for businesses to predict what will succeed or fail before introducing a new product. Demand planners discuss the anticipated demand for a future time through local meetings at various stages of the forecasting process. To develop calculations to support these operations, no historical information about product demand is used. The benefit of using this kind of forecasting for new product releases is that it works well.

Statistical forecasting based on historical data: Over the past decades, significant progress has been achieved in creating sales forecasting methods that take market factors into account. In assessments of sales forecasting procedures, however, only slight improvements in sales forecasting performance have been noted. Time series techniques are one method to use historical data as a forecasting input. In order to create a forecast, Open model Time Series (OMTS) algorithms evaluate the sales history and locate existing trends. The four data patterns that may be identified in the historical data are level, trend, seasonality, and noise. The forecast is based on how these patterns interact.

- Level is the pattern of sales that would occur in the absence of trends, seasonality, and noise.
- A trend is a recurring pattern of sales that might be upward or downward, linear or curved.
- A recurring pattern of sales growth or decline is known as seasonality.
- Noise is the random fluctuation that the time series approach cannot explain in the sales history.

Due to the complexity and low incremental accuracy, open time series approaches have not found widespread usage in industry. In contrast to open approaches, fixed time series techniques are employed by the studied company which is simpler since they make a priority assumption about the patterns present in the data (level, trend, seasonality, and noise). The demand planners will forecast the demand of the product considering historical data, trends, etc. But the quality of the data is one of the most important considerations in statistical forecasting.

iii. Joint forecast with customer collaboration: An essential component of the supply chain is the customer's participation in the demand planning process. Making a demand strategy and keeping track of it both assist. Data from a variety of sources, including social media, e-commerce analytics, and weather forecasts, are used in the consumer cooperation in the demand planning process. Considering this, a project named Joint forecast has been launched in the studied company with consumer involvement. The main objective of the project is customers providing their forecast inputs for the following three to six months. This will enable demand planners to adjust the forecast accurately and achieve a better supply fill rate. So, the demand planners can plan the production to ensure that there will be enough inventory to meet the demand of the customers by receiving forecast inputs for the future months and achieving the target of the company. This initiative stems from the realization that relying solely on historical data through statistical models is often insufficient, especially in cases where trends fluctuate or product histories are limited. By incorporating real-time customer inputs, demand planners can send timely signals to the supply and production teams to ensure enough inventory is available to meet future demand.

The Joint forecasting initiative also helps address external influences indirectly. By gathering real-time forecast inputs from customers, demand planners can adapt more quickly to external disruptions. For example:

- Customers may signal expected shifts in demand due to market disruptions or price changes.
- Their forecasts reflect the reality on the ground, allowing the company to adjust supply levels more accurately, even if external factors are at play.

Where external influences are known or expected (e.g., anticipated price hikes or major disruptions), demand planners are required to document assumptions upon which they base their adjustments. These assumptions provide a qualitative understanding of how external factors might affect future demand, complementing the statistical forecast.

4 Improvement and implementation outlines

The accountability, communication, and governance of the planning process have been highlighted as the main reasons that relate to this component of demand planning. Based on this, suggestions for prospective enhancements that emphasize market intelligence, conducting challenge-based consensus meetings, and utilizing direct input on the procedure might be developed. The expectations of the customers and the roles and duties of the associated stakeholders are made clearer. There is immediate information feedback on the approach during these sessions because of the process of analyzing and commenting on the prediction.

4.1 Forecast quality

Planning horizon, forecasting techniques, and product differentiation are significant fundamental factors that compromise prediction accuracy when it comes to forecast quality. To handle possible supply crisis, customers provide a six-month forecast horizon. The forecast, however, is subject to change each month, and the most recent forecast is considered. As a result, demand planners will find it simpler to plan production, ensure that there will be adequate stock in the inventory, and fulfill customer demand. Not all products should get equal forecasting attention. A distinction between product classes should be created since the profitability and revenue contribution of each product vary. The ABC classification of the products will be aided by the Pareto analysis. This strategy should be systemic, though. Additionally, it must be expanded upon and transformed into a proactive strategy. Where active forecasting based on product categories is used in addition to considering the past successes of these items.

The current circumstances of the joint forecast project are impacted by the prospective enhancements. The underlying factors causing the poor performance levels are within control, and uncertainty may be reduced. Each enhancement's impact on the demand planning process is shown in Table 1. The advantages and possible performance increase are shown in the last column. The contribution of an improvement to the primary objectives of the redesign will improve processes and reduce adverse forecast effects by eliminating uncertainty. Although this objective won't immediately improve forecast accuracy, it will lead to a more efficient process where uncertainty in the final demand plan is reduced in the long run.

4.2 Implementation in empirical domain

The implementation plan to get the best outcome for the prospective improvements is covered in this section. The largest impact on performance enhancements and higher fill rates appears to be the introduction of new implementations. Higher performances will result from these enhancements when coupled with additional organizational and procedural initiatives. One common template has been created and introduced to the customers where all customers must fill in their forecast inputs in this template and submit the forecast files to the demand planner via shared SharePoint© folders which are independent and limited to each customer. As soon as the customer enters the product's European Article Number (EAN) code, all additional details about that product will automatically fill in. The forecast months to choose from in the drop-down list (just one month) will automatically change beginning with the following months as the customer inputs the input date in the template.

Customers must therefore provide input for at least six months as required by the demand planners, but they can also provide input for a longer time horizon. The template is also designed to be safeguarded so that customers cannot alter any of the elements other than the area where they must submit their input. This is essential to prevent customers from making accidental errors/mistakes. To fill up the template, customers must choose their market name, enter the input date, choose the forecast months from the drop-down list which will automatically show the next month, and finally enter the list of EANs and quantities. The customers' focus will be only on the green colored cells/columns in the template.

Figure 9 provides an illustration of how the folders' connection is set up for the customers. The customers' names are not revealed in the figure and are just displayed as numbers due to privacy regulations. Customers must upload the forecast files in these folders each month; for instance, if they are submitting a forecast for the month of June, they must upload the file in the June folder. Demand planners can quickly and automatically trace who has submitted forecast inputs due to this method of submission. The entire process is automated with the aid of the Power Automate tool, and a flow is created so that whenever a file is uploaded in one of these SharePoint© folders, the tool can identify who uploaded it. For instance, if a file is uploaded in the December subfolder inside the folder 2, the tool can identify and

Table 1 Effects of improvements for joint forecast project

Improvements	Direct effect	Benefits
Standard template	Clarity of information Time consuming Better information alignment and management Pro-active approach towards Demand Planning process Less time consuming	More efficient process via the decrease of uncertainty
Tracking the customers who submitted the forecast inputs every month	Enhanced steering and controlling abilities Completely automated and no work for demand planners Less time consuming	More efficient process via the decrease of uncertainty
Creation of macro for consolidation process	Easy to consolidate large amount of data Less time consuming	More efficient process via the decrease of uncertainty
Dashboard creation for finding root cause	Less bias More flexibility Better concentrate on product groupings Added knowledge of the product life cycle Improved judgment on more important matters	Better forecasting More efficient process via the decrease of uncertainty

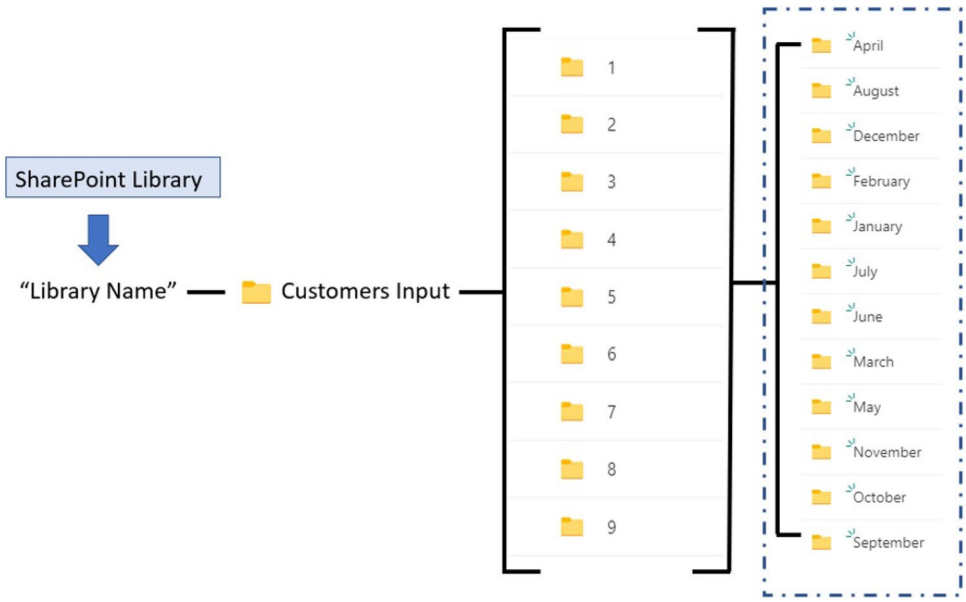


Fig. 9 The designed Customer's folder pattern joint prediction model

gives the information such as "customer 2 has submitted the file for December month", "date and time of the submission", and it extracts the information from the uploaded file and returns "which months the forecast is submitted for" in the excel tracker sheet.

To improve forecast accuracy, MAPE90 is a key indicator. The supply chain team and the sales department both use this indicator. The forecast is significant in the studied company organization because of the key and crucial function that demand planning plays. The production and distribution of the products are determined by the demand forecast. This indicator provides information on how well the product groups are performing. When forecasting is ineffective, it can substantially impair the system's performance, leading to large inventory levels, lower sales, expensive transportation, and uneven production levels. Forecast accuracy is a crucial factor in maintaining control in these scenarios. How closely the forecast matches the actual sales volumes is expressed by the forecast accuracy. This is accomplished by dividing the variation between the actual and anticipated sales levels by the latter. Once the forecast inputs and the actual orders are received, the demand planner need to find the deviation between forecast and actuals which gives MAPE90 results. Finding the MAPE90 is essential to find the root cause for the deviation. MAPE90 can be found at SKU level, house level, and subdivision level etc as defined at Eq. 1. The MAPE90 data will help demand planners identify which SKUs or houses have the highest deviations, which will help them increase forecast accuracy going forward. Dashboard can be used to find the top deviations caused at SKU or the house level.

$$MAPE90 = \frac{\sum_{EAN=0}^{n}([last3 - MonthActuals] - [last3 - MonthForecast])}{\sum_{EAN=0}^{n}[last3 - MonthActuals]} \quad (1)$$

A dashboard is created to find the root cause for the MAPE90 and visualize the data by graphs and charts. The dashboard contains calculating sheets, pivots, and charts. The actuals can be obtained from ERP system through Online Analytical Processing (OLAP) server which is directly connected to the Excel and forecast data can be obtained from customers' forecast input files located at SharePoint© folders. Based on the deviations, the root cause can be described by the demand planners, Root cause 1, Root cause 2, and Root cause 3 as illustrated in Table 2 are provided. The analytical tool is built on a foundation of both statistical and machine learning techniques. Key aspects include:

- Time Series Analysis: Algorithms identify patterns over time, which is essential for predictions based on historical data. The models consider trend and long-term movements in data and also seasonality as regular fluctuations occurring at specific intervals.
- Error Metrics: Metrics like MAPE90 are employed to evaluate forecast accuracy, helping identify biases and enabling continuous improvement.

Table 2 Root cause for MAPE90 schema

Root Cause 1	Root Cause 2	Root Cause 3
Initiatives	Launch phasing	Delayed or anticipated launch shipment for all or some customers
	Launch wrong sizing	Lower or higher launch volumes
	Promo phasing	Delayed or anticipated promotion shipment for all or some customers
	Promo wrong sizing	Lower or higher promotions volumes
	Promo cancelled	Cancelled by customerWrong phasing / calendarization - gift sets only
	Spring Sets incorrect phasing	
Retailer	Assortment issues	Discontinuation/assortment changes not reflected in retailer assortment
	Market conditions	Price change, weather, negotiations, market share change
	Event date changed	Delayed or anticipated event (ex : roll out implantation)
	Event volume changed	Lower or higher event volumes (ex : roll out implantation)
	Unusual order accepted	Retailer ordering more or less than its usual pattern
Forecasting	Stat Forecast not accurate	Wrong forecast calculation or Wrong seasonal model
	Cannibalization	Wrong cannibalization impact
	Phase-in/Phase-out	Same EAN forecast already on the new IRC while old IRC still ships
	Discontinuation	Discontinued IRC-no forecasts but actuals
	Manual forecast/Business Intelligence not accurate	Wrong manual forecast/business intelligence input
Supply	Out of Stock (OOS)	Out of stock in local warehouse not enough stock to cover retailer orders

The integration of data from JDA, ERP, and other systems into forecasting and supply chain management processes is a multifaceted effort. Through tools like MFR, OLAP, SQL databases, and automated Excel templates, the studied company not only streamlines its forecasting processes but also enhances decision-making capabilities. By effectively consolidating customer inputs and monitoring forecast accuracy, the supply chain efficiency and responsiveness to market demands can be improved.

4.2.1 Overall implementation roadmap

The overall plan for implementation consists of a 5-month plan. The main task can be proposed as:

1. *Planning*: The initial phase where project goals, scope, and deliverables are defined. This task likely spans the beginning of Month 1, indicating a foundational step in project development.
2. *Research*: Conducting necessary background research to inform subsequent steps. This task is also part of Month 1, suggesting its importance in setting the stage for effective implementation.
3. *Iterations 1 and 2*: These tasks represent phases of development and refinement. Iteration 1 occurs in Months 1 and 2, while Iteration 2 extends into Month 3, reflecting a cycle of testing, feedback, and adjustment.
4. *Testing*: An essential step where the developed components are evaluated for functionality and effectiveness. Testing is concentrated in Month 2, ensuring that any issues are identified early in the process.
5. *Internal Discussions*: This task, which occurs in Month 2, allows for team alignment and sharing of insights. Having two dedicated slots for discussion indicates an emphasis on communication within the team.
6. *Internal Discussion & Training*: A focused effort in Month 4 to ensure team members are well-equipped with knowledge and skills relevant to the project.
7. *Final Implementation*: The transition into the practical application of the project's outcomes. This task takes place in Month 4, showing that all preparatory steps are aligned towards execution.
8. *Documentation*: A critical task that ensures all aspects of the project are recorded for future reference. It is scheduled towards the end of Month 4, indicating a systematic approach to knowledge transfer.
9. *Introducing Standard Template to the Customer*: The final phase, taking place in Month 5, emphasizes the delivery of outcomes to the end user, which is crucial for stakeholder engagement and satisfaction.

These findings underscore the practical value of leveraging multiple data systems and automated tools to streamline demand planning and improve forecast accuracy. By consolidating customer inputs, firms can more reliably adjust production and inventory levels, thus reducing the risk of stockouts or excess stock. Demand planners become strategic enablers, using real-time analytics to refine models that might otherwise falter when historical patterns shift or become

insufficient. The chosen forecasting approach strikes a balance between predictive power and interpretability, ensuring that organizational stakeholders can both trust and act on the results. Ultimately, this integrated strategy supports greater responsiveness to fluctuating market demands, fostering a more resilient and competitive supply chain.

4.3 Discussions regarding the numerical analysis

In analyzing the research outcomes, it is imperative to acknowledge that they originate from numerical exemplifications anchored in explicit assumptions concerning market conditions and historical data patterns. External perturbations, encompassing a spectrum from geopolitical disturbances to fluctuations in raw material prices, may extend beyond the purview of purely statistical or time-series models unless they recur with sufficient frequency to be encapsulated within the underlying data. Consequently, certain potential inaccuracies or biases may emerge, particularly when atypical events or disruptions transpire that markedly deviate from established trends. To mitigate these risks, forecasting framework integrates human insights and real-time customer input alongside automated tools. This dual approach not only helps address the limitations inherent in purely data driven models but also enables faster adaptation to sudden market shifts or emerging disruptions. The Joint Forecasting Initiative, for example, provides immediate feedback from customers on expected demand changes, while business intelligence allows planners to manually adjust forecasts in response to external factors such as major supply chain issues or price hikes. Despite these mitigating measures, reliance on assumptions underscores the importance of ongoing model refinement and robust data collection, ensuring that forecasting processes remain agile, accurate, and resilient in the face of evolving market complexities.

4.4 Implications for research and practice

The results can be considered for exploring more robust frameworks that can seamlessly integrate both quantitative and qualitative insights, particularly for capturing unpredictable external factors. This includes refining hybrid or ensemble modeling techniques to incorporate sudden market shifts, while still leveraging manual intelligence to interpret unique or one-off disruptions. In practice, the evolution of automated tools and real-time data sharing should be used together with the development of flexible protocols, enabling continuous adjustments to plans as more sophisticated indicators become available.

5 Conclusion

This research findings provide potential information sources for demand planning as well as details on how to set up the process itself. The primary objective of this research is to enhance the current demand planning process in the Joint forecast project executed with customer participation. The ability to produce an acceptable degree of customer participation and customer service are examples of how a company performs. This dual goal is strongly related to inventory levels. Both excess and insufficient stock have an impact on these goals. Receiving customer forecast inputs and analyzing them to actually are the main issues with the Joint Forecast Project. This analysis will make it easier for the demand planners to create an accurate demand and properly establish the production level. Therefore, improving the demand planning process/tools boosts both organizational performance and customer forecast accuracy. Having a standard time horizon for submitting the forecast inputs for all the customers makes the process even better. By leveraging joint forecasting initiatives and implementing automated tools, managers can reduce operational risks through precise inventory alignment and support their strategic decision-making with accurate, real-time data. So, organizations not only improve supply chain agility and responsiveness but also create a sustainable framework for long term competitiveness and scalability. The analytical tool designed as a part of the Joint Forecasting Initiative represents an effective system that can integrate various forecasting methodologies and data sources, such as JDA and ERP systems. Through its robust architecture, algorithmic foundation, and seamless integration processes, the tool enhances the studied business case for forecasting capabilities, ultimately improving supply chain efficiency and responsiveness to market demands. The results of the implementation in the studied company have created potentials for the demand planning process with significant qualitative benefits like:

- Increased Efficiency: The automation of forecasting processes has substantially reduced the time and effort required from demand planners. By streamlining tasks such as data consolidation, analysis, and report generation, planners can

allocate their time more effectively to strategic activities, such as analyzing market trends and developing proactive supply chain strategies.
- Enhanced Accuracy: With automated tools minimizing manual errors, the forecasting process is expected to yield more accurate predictions. This improvement in forecast accuracy helps align inventory levels with actual demand, reducing the risk of overstocking or stock outs.
- Better Decision-Making: Automation facilitates real-time access to data and insights, empowering demand planners to make informed decisions more quickly. This responsiveness is crucial in a dynamic market environment, enabling company to adapt to changing consumer preferences and external market conditions.
- Employee Satisfaction: By reducing the manual workload and automating repetitive tasks, demand planners can focus on more meaningful and strategic aspects of their roles, potentially leading to increased job satisfaction and retention.

The demand planners may be able to identify the root cause of each deviation by determining the difference between the forecast inputs from the consumers and what the customers ordered. Demand planners may help customers by informing them of deviations, which will help them enhance the accuracy of their forecasts. Even the accuracy of the customer's forecast might be increased in this way. Potential oversimplifications or incomplete data attributes underscore the necessity for more robust data integration and interoperability of supply chain level systems besides the model sensitivity analysis to ensure any biases are minimized in forecasting initiatives.

Considering customer expectations plays a major role in creating a better demand strategy. Although this research demonstrates considerable advancements in customer input collection, the total impact of these advancements is still not entirely clear. Along with the process's direct consequences, there are other variables and influences that affect how accurately consumers can provide forecasts and how well they will respond to advancements already achieved. To have a more reasonable estimate of the possible organizational progress, it is advised to keep these additional elements in mind. It is helpful to extend the planning window or time fence. Therefore, it is advised that more study on this topic to assess the viability and usefulness of such a strategy will be conducted and to see whether the supply chain flexibility may be enhanced. Future studies could delve into advanced data analytics for unstructured data sources over supply chain operational systems, extended planning horizons, and collaborative forecasting platforms through supply chain stakeholders using emerging technologies like AI-driven modeling to improve the accuracy and agility of demand planning processes. Future research may also benefit from systematically tracking the impact of these integrated forecasting practices-through metrics such as cost savings, service-level improvements, and overall supply chain resilience-ultimately paving the way for more agile, data-driven decision-making processes.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and Consent to participate Not applicable.

Consent for publications Not applicable.

Competing interests The authors declare no Conflict of interest.

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